

Leo's Auto Database Project

Report 1

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Course and section: COSC 457 - 001

Instructor: Sung-Chul Hong Submission date: [date]

Contents

1 Introduction	2
2 Interview and target business	3
3 Business process	4
4 Data and application requirements	5
5 ERD schema and screen examples	6
6 Project schedule and meeting log	7
Appendix A Confirmed ERD and schema	[page]
Appendix B Completed interview form and signature ...	[page]

Who finishes each part

Owner	Deliverable
Interview Manager Mae	Introduction, interview, business process and requirements.
Database Manager [name]	Review Chen ERD and schema against the interview; attach the corrected design.
Application Manager [name]	Two or three screens, setup check, and any required changes.
Report Manager Dashaun	Combine the pieces, fill actual dates and contribution log, check pages and submit.

1 Introduction

Our project is a small database application that will use an interview with Leo's Auto Inc., the proposed business at 14 Willow Ave, Towson, MD 21286. The project focuses on the information needed to connect a customer, a vehicle, a repair visit, the work performed, and the resulting bill. We want to understand that process well enough to explain the data model and demonstrate a complete example using fictional records.

The purpose of the interview is to learn how the shop actually handles these activities. We will ask what information staff record, who supplies it, what happens during a normal visit, and which exceptions or reports matter. The shop's current tools and any difficulties with them will be described from its answers. We are not assuming that the shop uses paper records or needs to replace its existing software. [After the interview, add its date, the person's role, and two or three findings that shaped the project.]

Our proposed scope covers customer and vehicle records, repair orders, labor and parts used on a job, invoices, and payments. The initial reports are vehicle repair history, open repair jobs, and unpaid bills. These reports should let a user answer practical questions using related tables. Appointments, payroll, online payments, supplier ordering, and a customer portal are outside the proposed scope unless the interview or instructor makes one of them necessary.

The design phase will turn the interview answers into data and process requirements. A Chen entity relationship diagram will show the types of information and how they connect. A matching schema will identify the tables, primary keys and foreign keys. We will check rules such as whether a vehicle has one current customer, whether a repair can involve several mechanics, and whether one visit can produce more than one bill. Any difference between the interview and our starting design will be recorded and addressed.

The implementation phase will connect those tables to a small application. A Python and SQLite sample is available as a starting point, but our final tools will be [confirmed course tools]. We will use fictional records to test customer entry, a repair with labor and parts, invoicing, payment, and the selected reports. We will also check invalid entries and confirm that saved information remains available after restarting the application.

Work is divided among interview and requirements, database design, application development, and report assembly. Each member will review and explain their own contribution. The final package will include source code, database setup instructions, a functional guide, a step-by-step tutorial, and a demonstration. Our aim is a manageable application whose design, implemented behavior and documentation agree.

2 Interview and target business

Interview Manager:
Copy from the quick interview form

Interview record

Date and time: [fill]
Location or mode: [fill]
Interviewee and role: [fill]
Students present: [fill]
Signed proof: [attach the signed page as Appendix B]

The signature confirms that the interview occurred. It does not mean that the interviewee approved the report or application.

Business profile

Item	Interview answer
Business name and address	Proposed: Leo's Auto Inc., 14 Willow Ave, Towson, MD 21286. [Confirm]
History and branches	[Years operating, history, number of locations]
Size and staff	[Approximate staff size and relevant roles]
Activities and services	[Main services and daily business activities]
Customers and service area	[Who the shop serves and the area it covers]
Total sales	[Answer, or "not disclosed by interviewee"]
Current tools	[Software/forms used; what information is entered where]

Questions answers and evidence

Attach the completed interview form, including all questions and answers. If answers continue on another page, identify the question and add the continuation to Appendix B. Use the form headings as source labels in section 4.

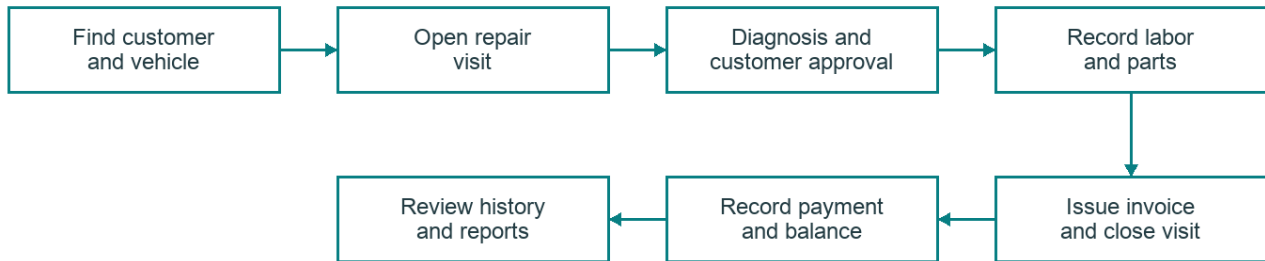
Main findings:

[1] _____
[2] _____
[3] _____

Still to confirm: _____

3 Business process

Interview Manager: Correct this sequence from the interview and identify who handles each step.



Proposed normal workflow.

Confirm the order of authorization, parts use, invoicing and payment during the interview.

One ordinary visit

[Describe an actual process from the interview: who accepts the customer, what is recorded, who approves work, how work and parts are added, how a bill is created, and how pickup/payment is handled. Do not include private customer details.]

Other activities in scope

Catalog and staff records: [who adds or updates parts, services and staff?]

Stock: [how parts are received, consumed and returned?]

Reports: [who needs each report and when?]

When something changes

Situation	What the shop does and what our app should do
Work not approved or cancelled	[Record the actual sequence, including stock already reserved/used.]
A part is unavailable	[Describe how the visit changes or stays open.]
Partial payment or billing correction	[Confirm deposits, refunds and invoice changes. The sample does not support all of these.]
Other important exception	[Example from the interview, or "none identified in scope".]

4 Data and application requirements

Interview Manager: Fill the answer column.

Database & App Managers: Change the design if an answer differs.

Interview topic and source	Data needed	Application behavior
Customer / vehicle fields Answer: [fill]	customer, vehicle Name/contact; VIN, year, make, model	Add/find customer and vehicle; avoid repeated VINs.
Job information / workflow Answer: [fill]	repair_order Complaint, diagnosis, odometer, authorization, status	Open a visit, authorize work, show open jobs, close or cancel.
Labor and parts Answer: [fill]	employee, service, labor_line, part, part_line	Record mechanic, service, hours, quantity and prices charged.
Stock handling Answer: [fill]	inventory_movement Part, quantity change, reason, optional job	Receive/use/return stock; prevent negative stock.
Billing rules Answer: [fill]	invoice, payment Bill amounts, payment amount, date and method	Issue bill; record allowed payments; calculate remaining balance.
Relationships Answer: [fill]	Vehicle ownership; mechanic per line; invoice per visit	[Confirm or revise the relationship counts in the ERD.]

Three proposed queries

Question	Input and expected result
What happened to this vehicle?	Filter by VIN; list repair dates, complaints and status, with billed totals when available.
Which jobs are still open?	No input; show open/in-progress orders with vehicle/customer and opening date.
Which bills are still unpaid?	No input; show invoices with total, amount paid and positive remaining balance.

Interviewee priorities: [keep / change the three reports above, with reasons].

Other required fields or functions: [add anything missing].

Unanswered questions and person following up: [fill].

Check these answers against the fields and rules in Appendix A.

5 ERD schema and screen examples

Conceptual design and schema

Database Manager:

check Database_Guide.pdf against the interview and attach it as Appendix A. Keep both Chen sheets together. Editable diagrams and schema are in the diagram_sources folder beside this report.

Our proposed design contains 11 entities. A customer can have several vehicles, and each vehicle can have several repair visits. Each visit records its bill-to customer separately. Labor and parts are stored as individual lines so a visit can include several services or parts. Each line records what was charged at the time. An invoice belongs to one repair visit, and several payments may apply to that invoice.

Changes made after the interview: [list each changed field, relationship or rule and its reason].

Unchanged assumptions checked with the interviewee: [list].

Keys constraints and normalization

Primary keys identify individual records. Foreign keys connect records without copying the full customer, vehicle or part into each row. A unique VIN avoids duplicate vehicle records, and a unique order link allows at most one invoice per visit in the proposed model. Stock and payment checks prevent negative stock and overpayment. Historical job prices and invoice labels are preserved so later catalog changes do not rewrite old bills.

[Confirm these rules match the interview. Explain any rule removed or changed. State the selected database engine and how its implementation will enforce the rules.]

Application screens

Application Manager: add two or three screenshots or screen sketches here.

Screen	Input and output to show
Customer and vehicle	Input: name/contact and VIN/car details. Output: saved linked records or an error. [Insert screenshot/sketch]
Repair and billing	Input: complaint, authorization, labor and parts. Output: repair lines, invoice total and payment balance. [Insert screenshot/sketch]
Report	Input: report choice / vehicle filter. Output: readable results for one of the three queries. [Insert screenshot/sketch]

Required tools: [fill]. Sample currently uses Python and SQLite.

Changes needed before implementation: [fill].

6 Project schedule and meeting log

Report Manager:

Fill real calendar dates and names. The shaded cells below suggest the order of work; move them to fit the actual deadlines.

Report 1 due: [date] Final due: [date]

A [dates]: interview B [dates]: Report 1 C [dates]: build D [dates]: final

Task	Owner	A	B	C	D
Interview and report content	Mae	X	X		
ERD schema and report assembly	DB + Report		X		
Database and app integration	DB + App			X	
Full checks and fixes	Everyone			X	X
Manuals slides and rehearsal	Report + all				X

Concrete handoffs

Interview Manager posts the completed form and report sections by [date].

Database Manager posts the checked diagram/schema by [date].

Application Manager posts screens and setup notes by [date].

Report Manager posts the assembled draft for everyone to review by [date].

Actual meeting and contribution log

Record real meetings or group-chat work: date, place/channel, each person's contribution and next task.

Date time and place or channel	Each person's actual contribution	Decision and next step
[Fill after it happens]	Mae: [fill] Marquise: [fill] Marcel: [fill] Dashaun: [fill]	[What was agreed; task, owner and due date]
[Next actual session]	[Names and contributions]	[Decision and next task]

Before submitting

- [] Interview answers and genuine signature attached
- [] All placeholders resolved and introduction updated
- [] Processes, requirements, ERD and schema agree
- [] Screens, dated schedule and accurate log included
- [] Everyone checked their section; contents/page numbers updated